CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

November 2013

EE416 Advanced Microcontrollers & Applications

Time: 10.00am to 01.00pm **Date: 23-11-2013** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. **Comments are mandatory with each line of code.**

SECTION – I

Q-1	(a)	1.	Give two reasons for faster execution speed of c8051f120 microcontroller with respect to Intel 8051 microcontroller.	2
		2.	Draw & Explain memory architecture of 8051 microcontroller.	5
Q-2	(a)	1.	List out various peripherals available in c8051f120 microcontroller.	4
		2.	Write a program to initialize c8051f120 microcontroller for 98 MHz internal clock with watch-dog disable & all port pins functioning in Push pull mode.	5
		3.	Write & explain a program which loads an integer number into TH0:TL0 using any two methods.	5
			OR	
	(a)	1.	A microcontroller (c8051f120) program is communicating with external device via UART, Producing PWM signal only on CEX2 & CEX3 & using AIN2.0 & AIN2.1 for analog input. Find out Pin assignment as per their priorities mentioned in Crossbar decoder.	4
		2.	Compare C language & Assembly programming method for development of an Embedded application.	5
		3.	Write & explain a program which demonstrates use of structure & union.	5
Q-3	(a)	1.	Explain the role of crossbar in c8051f120.	4
		2.	Explain working of Timer 2 in 16 bit Auto reload mode with necessary diagram. List out the SFR used for the same.	5
		3.	Explain I2C protocol. Explain how master read from or write into slave device with necessary waveforms.	5
			OR	
	(a)	1.	Explain use of various control loops for C programming of microcontroller.	4

		2.	Explain working of 16 bit PWM mode available as a part of PCA module in c8051f120.	5
		3.	Calculate the number to be loaded in various register for above question if required PWM frequency is 18 KHz with a duty cycle of 20 %.	5

SECTION – II

Q-4	(a)	1.	Explain following RTOS functions with example program. 1. Os_create( ) 2. Os_delete( )	2
		2.	Calculate the period for time slice for default setting available in CONF_TNY.A51 file for a crystal frequency of 10 MHz, 12 clock mode.	5
Q-5	(a)	1.	Write a program which measures signal available on AIN0.0 with sampling time of 1ms. The sampling should be initiated on overflow of timer 2. Show necessary calculations.	6
		2.	Write a program to measure bi-polar signal available on channel 0 of ADC2 (AIN2.0 & AIN2.1) with SAR frequency of 1.5 MHz. The conversion should begin on writing to AD0BUSY bit. Show necessary calculations.	8
			OR	
	(a)	1.	Explain working of DAC available in c8051f120.	6
		2.	Generate a half wave rectified signal on DAC0 pin using an array of 20 data. The DAC update should occur by writing to DAC0H register. Show necessary calculations to decide the value used in array.	8
Q-6	(a)	1.	Explain LCD interfacing with c8051f120 microcontroller. Draw necessary hardware interfacing diagram.	6
		2.	Write a program for single phase over current relay. The relay samples the signal by ADC0 at rate of 1ms & produces a trip signal on pin P1.3.	8
			OR	
	(a)	1.	Draw hardware diagram for closed loop speed control of DC motor. Also list out the peripherals (of c8051f120 microcontroller) required to be used for such control.	6
		2.	Write & explain a program for close loop speed control of DC motor.	8